



Siddharth Simediya

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Personal Data

Github: <https://github.com/Siddharth2412>

Professional Profile

Engineer with hands-on experience integrating AI capabilities into production systems and building API-driven automation: from a GWDG AI chat platform (image generation, audio transcription, memory integration, health-checked AI models) to a self-built FastAPI/Pydantic booking platform with REST APIs and TOTP-based authentication. I automate workflows with Python and browser/API scripting, model business rules directly into validation logic, and ship changes through CI/CD pipelines and Docker. I want to bring that AI-integration and automation background to building and operating AI-driven automation solutions.

Professional Experience

Sep 2023 - Sep 2025 **Research Assistant, GWDG** (*Gesellschaft für wissenschaftliche Datenverarbeitung mbH*), Göttingen, Germany

Chat AI project

- Integrated image generation, audio transcription, and memory functions into an AI chat platform
- Implemented health-check mechanisms for AI models running on HPC systems
- Developed server-side tools for job and service management on HPC/cloud-native platforms; optimized the scheduler to prevent system crashes
- Introduced a scale-to-zero approach to reduce GPU resource usage, balancing cost against availability

Ironik project

- Automated validation of Kubernetes cluster configurations against defined schemas, improving system stability
 - Used Ansible to manage and configure Rancher-based Kubernetes clusters
- Built monitoring and observability tooling across all projects using Prometheus and Grafana

Aug 2022 - Oct 2023 **Werkstudent Python Developer, Digital Media 360**, Munich, Germany

- Built web scraping and workflow-automation tools with BeautifulSoup and Selenium
- Backend development in Python, applying behavior-driven testing (BDD) methods; ran CI workflows with Jira and GitLab

Jan 2018 - Dec 2020 **E-commerce Functional Executive, Alpyog Marketing**, Surat, India

- Developed web scraping software for e-commerce platforms, integrated with Google Sheets via the Google Cloud API
- Automated manual inventory processes through custom software solutions; implemented CSV validation and automated form submission
- Built backend data processing in Python and implemented REST APIs
- Automated Google Sheets workflows using Google Apps Script and JavaScript

Independent Project

Since Mar 2026

Kinderkreis (Demo): Full-Stack Booking Platform for Certified Child-care

- Built a FastAPI/Pydantic v2 REST API (20 endpoints) with two independent authentication systems: email-OTP user sessions and TOTP-based two-factor admin sessions
- Modeled real regulatory business rules (childcare ratios, certification thresholds) directly into Pydantic validation, with server-side error handling and automatic .ics calendar invites
- Wrote a 137-test pytest suite with isolated test databases per case; deployed via GitHub Actions CI/CD (self-hosted runner) to a production VM with Docker Compose
- **GitHub:** <https://github.com/Siddharth2412/kinderkreis-demo>

Education

Oct 2022 - Mar 2026

M.Sc. Applied Computer Science, *Georg-August-Universität Göttingen*, Göttingen, Germany

Aug 2016 - Aug 2020

B.E. Information Technology, *Sarvajani College of Engineering & Technology*, Surat, India

Technical Skills

Programming & Scripting

Python (Advanced), JavaScript, Bash, PowerShell

APIs & Backend

FastAPI, REST APIs, Pydantic v2, TOTP 2FA authentication

Automation & Scripting

Selenium, BeautifulSoup, Google Apps Script, CSV/form automation

CI/CD & Containers

GitLab CI, GitHub Actions, Docker, Docker Compose, Git

AI Platform Integration

AI chat platform feature integration, AI model health-checking on HPC systems

Monitoring & Observability

Prometheus, Grafana

Languages

English

C1 - Professional working proficiency

German

B2 - Upper-intermediate